

|          |           |        |                       |            |                                                   |
|----------|-----------|--------|-----------------------|------------|---------------------------------------------------|
| $P$      | price     | $C(Q)$ | total cost            | $Q^*, P^*$ | profit-maximizing quantity and price              |
| $Q$      | quantity  | $F$    | fixed cost            | $P_0$      | price at which nobody buys                        |
| $R$      | revenue   | $c$    | cost of one more unit | $Q_F$      | quantity where demand meets MC                    |
| $\Delta$ | change in | WTP    | willingness to pay    | slope      | $\Delta P / \Delta Q$ , slope of the demand curve |

## Decisions and Trade

|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| Net benefit                  | benefit - direct cost                                        |
| Opportunity cost             | net benefit of the next best alternative                     |
| Economic cost                | direct cost + opportunity cost                               |
| Economic rent                | net benefit - opportunity cost                               |
| Opportunity cost of one unit | $\frac{\text{other good given up}}{\text{this good gained}}$ |

## Costs

|               |                                                  |
|---------------|--------------------------------------------------|
| Average cost  | $AC = \frac{C(Q)}{Q}$                            |
| Marginal cost | $MC = \frac{\Delta C}{\Delta Q} = C(Q+1) - C(Q)$ |

## Elasticity

|                        |                                                                                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Percentage change      | $\% \Delta = \frac{\text{new} - \text{old}}{\text{old}} \times 100$                                                                                                      |
| Price elasticity       | $\epsilon = -\frac{\% \Delta \text{ demand}}{\% \Delta \text{ price}} = -\frac{P}{Q} \times \frac{\Delta Q}{\Delta P}$<br>$= -\frac{P}{Q} \times \frac{1}{\text{slope}}$ |
| Income elasticity      | $\epsilon_{\text{income}} = \frac{\% \Delta \text{ demand}}{\% \Delta \text{ income}}$                                                                                   |
| Cross-price elasticity | $\epsilon_{\text{cross}} = \frac{\% \Delta \text{ demand for A}}{\% \Delta \text{ price of B}}$                                                                          |
| Markup rule            | $\frac{P - MC}{P} = \frac{1}{\epsilon}$                                                                                                                                  |

## Revenue and Profit

|                  |                                                  |
|------------------|--------------------------------------------------|
| Revenue          | $R = P \times Q$                                 |
| Profit           | $R - C(Q)$                                       |
| Marginal revenue | $MR = \frac{\Delta R}{\Delta Q} = R(Q+1) - R(Q)$ |
| Profit maximum   | $MR = MC$                                        |

## Surplus

### On one sale

|               |            |
|---------------|------------|
| Buyer's gain  | $WTP - P$  |
| Firm's gain   | $P - MC$   |
| Joint surplus | $WTP - MC$ |

### In the market

|                  |                                                          |
|------------------|----------------------------------------------------------|
| Consumer surplus | $CS = \frac{1}{2} \times Q^* \times (P_0 - P^*)$         |
| Producer surplus | $PS = (P^* - MC) \times Q^*$                             |
| Deadweight loss  | $DWL = \frac{1}{2} \times (Q_F - Q^*) \times (P^* - MC)$ |

### Areas

|           |                                                       |
|-----------|-------------------------------------------------------|
| Triangle  | $\frac{1}{2} \times \text{base} \times \text{height}$ |
| Rectangle | $\text{base} \times \text{height}$                    |

